

4DW-VLA: 4D and World-Model Aligned Vision-Language-Action for Robotic Manipulation

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Abstract—Most vision-language-action (VLA) policies operate on 2D images, yet robots act in a metric 3D workspace whose configuration evolves over time—a 4D problem. Methods that inject 3D or 4D structure often rely on depth sensors, point-cloud encoders, or heavy scene reconstruction, which increases model size, training cost, and deployment latency; many still treat spatial geometry and temporal dynamics as separate concerns. World-model-augmented VLAs predict future 2D video but rarely align that visual foresight with the robot’s own metric body motion. We present 4DW-VLA, a 4D and world-model aligned VLA that jointly models (i) future 4D body trajectories via lightweight FK and (ii) future 2D egocentric video, and uses both to condition chunked action generation. A 4D Future Expert and World Knowledge Expert exchange information through Unified Bidirectional Cross-Attention; a frozen large video world model supervises training only and is removed at inference. On three RoboTwin 2.0 tasks, 4DW-VLA matches or exceeds representative 2D-foresight and depth-plus-video baselines, with the largest gains under domain randomization.

I. INTRODUCTION

Learning-based manipulation policies and VLA models [2, 3, 5, 29, 11, 1] have improved language following and cross-task generalization, but they still under-represent the physical world in which a robot moves.

Most VLAs are fundamentally 2D. They consume RGB images and language, yet manipulation unfolds in a 3D workspace; when time is included, the relevant object is **4D** structure—how 3D geometry changes across the action horizon. A policy that reasons only in image space lacks explicit metric body motion and cannot guarantee that predicted actions respect the robot’s own spatial evolution.

3D-/4D-aware VLAs often pay a heavy geometric tax. Point clouds, depth maps, voxel grids, and 3D Gaussian fields can improve spatial reasoning [27, 26, 6], but they typically require extra sensors or large perception networks to extract and encode geometry at every timestep. The resulting models are bigger, slower to train, and harder to deploy. Moreover, many 3D-enhanced designs treat geometry as a **static** per-frame cue and under-model **temporal** changes in the robot’s own configuration.

Temporal prediction is usually misaligned across modalities. Actions are inherently future-oriented: a policy outputs action chunks over horizon H . Yet most VLAs do not jointly predict how **2D observations** and **3D/4D body state** should

co-evolve over that horizon. World-model and world-action-model (WAM) lines predict future video [7, 14, 18], but often without explicit metric 4D body motion; the camera may “imagine” a future that is not consistent with the robot’s actual 3D trajectory (e.g., the arm rises in joint space but the predicted video does not show the raised hand).

These signals should be coupled. When a robot lifts its end-effector in 3D, the egocentric camera stream should reflect that motion; conversely, visual foresight should not contradict feasible body dynamics. Effective manipulation therefore requires **aligning** (a) future **4D body motion** in the robot frame and (b) future **2D video** the camera will see, and using both to predict (c) future **actions**.

We propose **4DW-VLA—4D and World-model aligned VLA**—to address the above gaps. Our design rests on three ideas:

- 1) **Low-cost 4D from FK.** We compute 7D link trajectories (3D position + orientation) directly from proprioception via forward kinematics (FK), avoiding the orders-of-magnitude overhead of perception-based 4D pipelines.
- 2) **Unified Bidirectional Cross-Attention (UBCA)** between a **4D Future Expert** and a **World Knowledge Expert**, coupling 4D body foresight and 2D visual foresight before action decoding.
- 3) **Train with a frozen world model; deploy without it.** A compact expert distills a frozen large video world model during training; the world model is discarded at inference.

Fig. 1 summarizes the architecture. We co-train four objectives under one weighted multi-task loss (Section III-E). The video world model g_{wm} stays **frozen**; the VLM backbone is fine-tuned at **one-fifth** the learning rate of the expert and action heads over **10,000** optimizer steps.

Contributions:

- 1) Formulation of **4D–2D–action alignment** for VLA manipulation (Section III-A).
- 2) A **4D Future Expert** using **FK-derived 7D link trajectories** (Section III-B).
- 3) **UBCA** between the 4D and World Knowledge experts (Section III-C).
- 4) Distillation of a **frozen** video world model with **no deployment-time generation** (Sections III-C–III-F).

- 5) A **multi-task training objective** with tunable weights (Section III-E).

II. RELATED WORK

A. Foundation Models in Robotics

Generalist robot policies are trained on large multi-task datasets [25, 10, 24] and increasingly built on vision-language priors [3, 11, 1, 13]. RT-2 [3] and OpenVLA [11] cast actions as tokens; Diffusion Policy [5], RDT [13], and π_0 [1] generate continuous action chunks. These foundations excel at semantic generalization but largely treat observations as 2D images without explicit 4D body–video alignment.

WAMs co-train video prediction with control [14, 18, 16]. **4DW-VLA** follows the “train with foresight, infer without the generator” principle, but explicitly couples **FK-derived 4D body trajectories** with **2D video foresight** through UBCA before action decoding.

B. 3D and 4D Representations for Manipulation

Methods that inject 3D or 4D structure improve spatial and temporal reasoning, but the **4D signals they use are typically heavy**—extra sensing, large encoders, auxiliary foundation models, or scene-scale reconstruction inflate compute, memory, and model size. **4DW-VLA** derives 4D body motion from FK and distills 2D dynamics into a small expert.

RynnWorld-4D (Alibaba DAMO Academy) [28] co-generates future RGB, depth, and optical flow via tri-branch diffusion on hundreds of millions of frames—**infrastructure-heavy**. **4D-VLA** [22] injects sequential RGB-D with 3D coordinate embeddings; **depth must be available or estimated for every frame**. **Pri4R** (LG AI Research) [20] uses privileged scene point tracks from depth-based tracking—**expensive to obtain and encode**. **Lift3D-VLA** [19] lifts VLAs with explicit point clouds and GC-MAE pretraining—**substantial 3D encoder overhead**. **SpatialVLA** [15] integrates monocular depth (ZoeDepth) into a 4B-scale VLA—a **permanent depth branch** at inference unless distilled away. **4D-WAM** [17] aligns WAMs with dense pixel-level 3D trajectory fields from a large 4D teacher.

In contrast, **4DW-VLA** obtains 4D through **FK on proprioception** and aligns it to 2D foresight through **two small coupled experts**, using a frozen world model only as a **training-time teacher**.

C. World Models and World-Action Models

Classical world models predict future states [7, 12, 8]. Recent WAMs make the future action-facing [14, 21]. **4DW-VLA** aligns future pixels with future FK 4D body motion via UBCA, then conditions actions on both distilled pathways.

III. METHOD

We present 4DW-VLA, a 4D and world-model aligned VLA for language-conditioned manipulation. A VLM encodes multi-view RGB and instructions; three trainable heads—a **4D Future Expert**, a **World Knowledge Expert**, and an **Action Expert**—cooperate through UBCA. Fig. 1 shows the overview.

A. Problem Formulation

Let o_t denote multi-view RGB, ℓ_t the language instruction, and q_t proprioception. Write p_t for the current metric 3D body state (FK-derived link poses), $P_{t+1:t+H}$ for the future 4D body trajectory, and $V_{t+1:t+T}$ for future egocentric video.

4D–2D future alignment:

$$p(P_{t+1:t+H}, V_{t+1:t+T} \mid o_t, \ell_t, q_t). \quad (1)$$

Action prediction:

$$p(A_t \mid P_{t+1:t+H}, V_{t+1:t+T}, o_t, \ell_t, q_t), \quad A_t = [a_t, \dots, a_{t+H-1}]. \quad (2)$$

At deployment we use compact representations Z_t^{4D} and Z_t^{wk} :

$$p(A_t \mid Z_t^{4D}, Z_t^{wk}, o_t, \ell_t, q_t). \quad (3)$$

Current 3D state estimation (auxiliary):

$$p(p_t \mid o_t, \ell_t, P_t^{\text{hist}}), \quad (4)$$

where P_t^{hist} is a short history of FK 4D tracks.

B. 4D Future Expert

Manipulators expose q_t ; FK maps joints to link poses in microseconds. For each link/keypoint $k \in \{1, \dots, K\}$,

$$x_t^k = \text{FK}_k(q_t) = [p_t^k, r_t^k] \in \mathbb{R}^7, \quad (5)$$

where $p_t^k \in \mathbb{R}^3$ is position and $r_t^k \in \mathbb{R}^4$ orientation (unit quaternion) in the robot base frame.

The 4D Future Expert is a lightweight transformer over temporal 4D tokens from q_t and P_t^{hist} , conditioned on the VLM prefix. It outputs reconstructed current 4D state $\hat{P}_t$, predicted future trajectory $\hat{P}_{t+1:t+H}$, and hidden tokens Z_t^{4D} .

Proprioception q_t anchors predictions and, through UBCA, supplies metric body context to the 2D foresight pathway.

4D future loss:

$$\mathcal{L}_{4D} = \beta \left(\text{MSE}(\hat{P}_t(q_t), P_t) + \gamma \text{MSE}(\hat{P}_{t+1:t+H}(q_t), P_{t+1:t+H}) \right). \quad (6)$$

Current state loss:

$$\mathcal{L}_{\text{cur}} = \text{MSE}(\hat{p}_t, p_t). \quad (7)$$

C. World Knowledge Expert and 4D–2D Alignment

The **World Knowledge Expert** learns implicit 2D-foresight tokens $Z_t^{\text{wk}} \in \mathbb{R}^{M \times d}$ aligned with the 4D Future Expert.

UBCA. At layer ℓ and timestep i , for module $A \in \{4D, \text{wk}\}$ with companion B :

$$\tilde{h}_{\ell,i}^A = \text{Attn}(q(h_{\ell,i}^A), K_{\leq i}^A, V_{\leq i}^A) + \text{Attn}(q(h_{\ell,i}^A), K_{\leq i}^B, V_{\leq i}^B), \quad (8)$$

with causal masking over timesteps. Fig. 2 shows the structured attention pattern.

The world-knowledge pathway produces

$$Z_t^{\text{wk}} = f_{\text{wk}}(o_t, \ell_t, q_t, Z_t^{4D}). \quad (9)$$

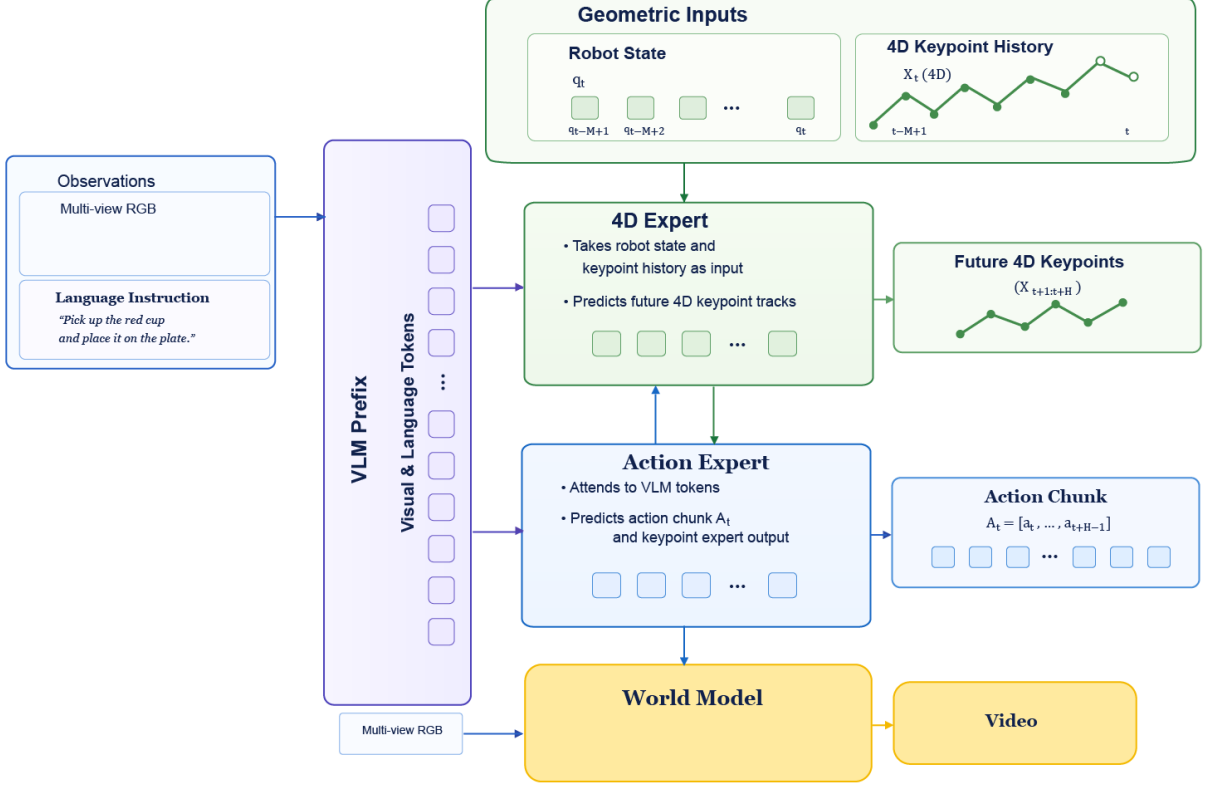


Fig. 1: **4DW-VLA overview.** A VLM prefix encodes RGB and language. The **4D Future Expert** predicts current and future FK 7D link trajectories. The **World Knowledge Expert** predicts implicit 2D-foresight tokens aligned with 4D through UBCA. A **frozen** video world model supervises 2D foresight during training only. The **Action Expert** consumes VLM tokens and both experts’ outputs. At inference, the world model is dropped.

During training, Z_t^{wk} conditions a **frozen** video diffusion transformer g_{wm} . With VAE latents x_0 , noise ϵ , and $\tau \sim \mathcal{U}[0, 1]$:

$$\mathcal{L}_{\text{vid}} = \mathbb{E}_{\tau, \epsilon} \left[\|g_{\text{wm}}(x_\tau, h(Z_t^{\text{wk}})) - (\epsilon - x_0)\|^2 \right]. \quad (10)$$

Only f_{wk} , UBCA parameters, and projection h are updated; g_{wm} is **frozen**.

D. Action Expert

The Action Expert generates action chunk A_t via flow matching, conditioned on the VLM prefix, Z_t^{4D} , Z_t^{wk} , and q_t . With ground-truth A_t^0 , noise ϵ , and $\tau \sim \mathcal{U}[0, 1]$:

$$\mathcal{L}_{\text{act}} = \mathbb{E}_{\tau, \epsilon} \left[\|v_\theta(A_t^\tau, o_t, \ell_t, q_t, Z_t^{4D}, Z_t^{\text{wk}}) - (\epsilon - A_t^0)\|^2 \right]. \quad (11)$$

E. Multi-Task Training Objective

$$\mathcal{L} = \lambda_{\text{cur}} \mathcal{L}_{\text{cur}} + \lambda_{4D} \mathcal{L}_{4D} + \lambda_{\text{vid}} \mathcal{L}_{\text{vid}} + \lambda_{\text{act}} \mathcal{L}_{\text{act}} + \lambda_{\text{vqa}} \mathcal{L}_{\text{vqa}}. \quad (12)$$

Post-training runs for **10,000** steps. The video world model g_{wm} is **frozen**. All other trainable modules—VLM, 4D Future

Expert, World Knowledge Expert, UBCA, projection h , and Action Expert—are updated jointly from step 0. Let η denote the base learning rate for experts and the action head; the VLM uses $\eta/5$. No separate warmup stage is applied.

F. Inference

At deployment: (1) encode (o_t, ℓ_t) with the VLM; (2) run both experts with UBCA to obtain Z_t^{4D} , Z_t^{wk} ; (3) run the Action Expert to output A_t ; (4) **do not** invoke g_{wm} . The extra cost vs. a plain 2D VLA is two lightweight expert forward passes and FK on q_t .

IV. EXPERIMENTS

We evaluate 4DW-VLA on RoboTwin 2.0 [4] against two **representative baseline families**:

- 1) Does explicit **4D–2D alignment** improve Easy success over a **2D-foresight** VLA?
- 2) Does the recipe improve Hard (OOD) robustness?
- 3) Do gains persist across stacking, tool-use scanning, and placement?

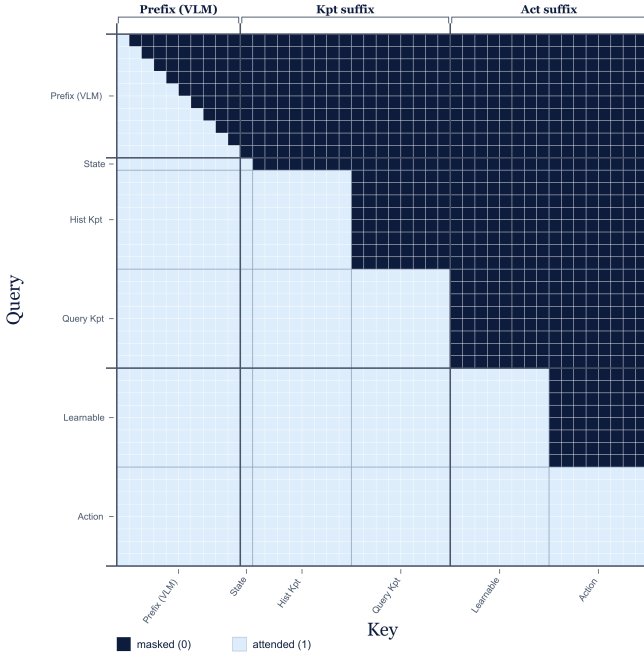


Fig. 2: **Attention structure.** Within the coupled 4D / World Knowledge block, tokens attend causally within their stream and to causal history in the partner stream.

A. Experimental Setup

Benchmark. RoboTwin 2.0 uses SAPIEN with domain randomization. Easy = demo_clean; Hard = demo_randomized. Aloha-AgileX bimanual embodiment; success rate is the metric.

Tasks. stack_bowls_three, scan_object, place_bread_skillet.

Baselines. Neither baseline is an architectural parent of 4DW-VLA.

InternVLA-A1.5 [9] represents the **2D-foresight VLA** family: temporal reasoning and future **2D video** distillation, but **no explicit 3D/4D body motion** or 4D-2D alignment. We re-implement and train it under the same budget as 4DW-VLA.

LingBot-VLA 2.0 [23] represents the **depth-plus-video VLA** family: **3D (depth)** cues and temporal video generation, but **no coupling** of 3D/4D body trajectories with 2D temporal evolution. Numbers are **as-reported**; we do not reproduce it. Not evaluated on scan_object or place_bread_skillet (n/a).

Training recipe. 4DW-VLA and re-implemented InternVLA-A1.5 share **10,000 steps**, frozen g_{wm} , VLM at $\eta/5$, experts/action at η , no warmup (Table II).

B. Main Results

Fig. 3 visualizes Table I.

stack_bowls_three. 4DW-VLA reaches 81%/58% (Easy/Hard) vs. InternVLA-A1.5 (71%/56%) and LingBot-VLA 2.0 (78%/22%). On Hard, LingBot-VLA 2.0 drops to

22% while 4DW-VLA holds 58%—consistent with **4D-2D aligned body foresight** under visual randomization.

scan_object. 4DW-VLA ties InternVLA-A1.5 on Easy (45%) and leads on Hard (33% vs. 28%).

place_bread_skillet. 4DW-VLA reaches 39%/29% vs. 31%/20% for InternVLA-A1.5 (+8/+9 points).

Across evaluated columns, 4DW-VLA is best reported. Hard gains over InternVLA-A1.5: +2, +5, +9 points on the three tasks.

C. Discussion

Easy scores match or exceed InternVLA-A1.5 on all tasks; consistent Hard improvements support **explicit 4D-2D coupling** over **2D video foresight alone**. UBCA and individual loss contributions are not yet ablated. LingBot-VLA 2.0 numbers are unverified external references.

D. Real-Robot Validation (In Progress)

Real-robot evaluation (4DW-VLA vs. InternVLA-A1.5) is in progress; protocol and results will be reported before final submission.

V. LIMITATIONS

FK trajectories capture body motion, not independent object dynamics. URDF-hardware mismatch propagates into 4D labels. Removing g_{wm} at inference assumes Z_t^{wk} captures sufficient visual dynamics. UBCA ablations and loss-component isolations are pending. Only three RoboTwin 2.0 tasks are reported; trial counts and confidence intervals remain to be finalized.

VI. CONCLUSION

We presented **4DW-VLA**, a **4D and world-model aligned VLA**. It uses lightweight FK 7D trajectories, UBCA to align future 4D body motion with future 2D video, and distills a frozen video world model at training time while discarding it at inference. On RoboTwin 2.0, 4DW-VLA matches or exceeds a **2D-foresight** representative (InternVLA-A1.5) and a **depth-plus-video** representative (LingBot-VLA 2.0, as-reported), with the strongest advantage under domain randomization.

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TABLE I: **Post-training success rates (%) on RoboTwin 2.0.** Easy = demo_clean; Hard = demo_randomized. Bold = best per column.

Method	stack_bowls_three		scan_object		place_bread_skillet	
	Easy	Hard	Easy	Hard	Easy	Hard
InternVLA-A1.5	71	56	45	28	31	20
LingBot-VLA 2.0	78	22	n/a	n/a	n/a	n/a
4DW-VLA (ours)	81	58	45	33	39	29

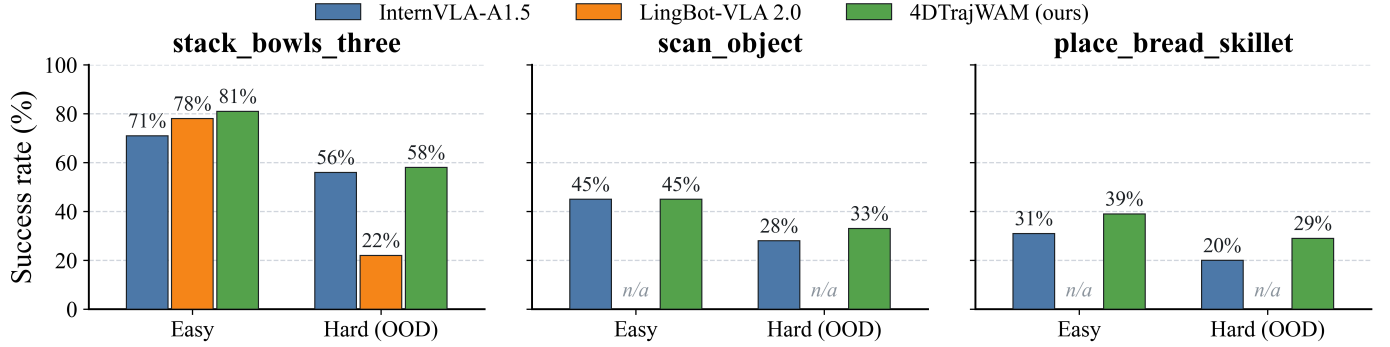


Fig. 3: **RoboTwin 2.0 success rates.** Grouped bars for Easy and Hard on three tasks. LingBot-VLA 2.0: n/a on two tasks.

TABLE II: **Shared post-training recipe (4DW-VLA and re-implemented InternVLA-A1.5).**

Item	Setting
Benchmark	RoboTwin 2.0
Post-training budget	10,000 steps
Video world model g_{wm}	Frozen
VLM learning rate	$\eta/5$ (experts / action at η)
Warmup	None
Easy / Hard	demo_clean / demo_randomized

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